

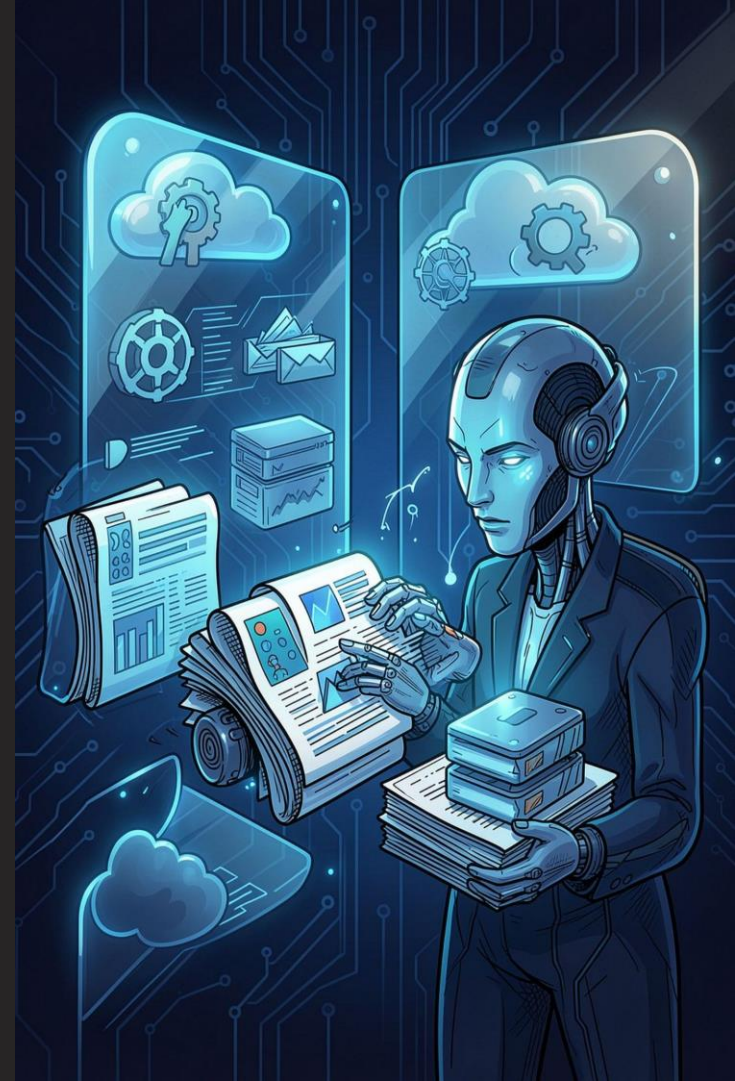
AI Research Assistant Platform

AI Powered Research & Knowledge Management

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COMMON RESEARCH CHALLENGES

SCATTED PAPERS



Problem Statement

Researchers waste substantial time on discovery, organization, and tracking. The core issues:

- Time-consuming manual literature search and reading
- Difficulty managing and syncing references
- Information overload — hard to keep up with new publications
- No centralized, searchable repository for research assets
- Missed or late discovery of relevant papers



Objectives



Automate collection

Continuous literature harvesting from top sources



AI paper summarization

Concise, structured summaries for fast review



Smart keyword extraction

Context-aware terms and concept tagging



Efficient reference management

Organize citations and PDFs automatically



Searchable repository

Full-text and metadata search across saved items



Weekly research digest

Curated highlights delivered to inbox

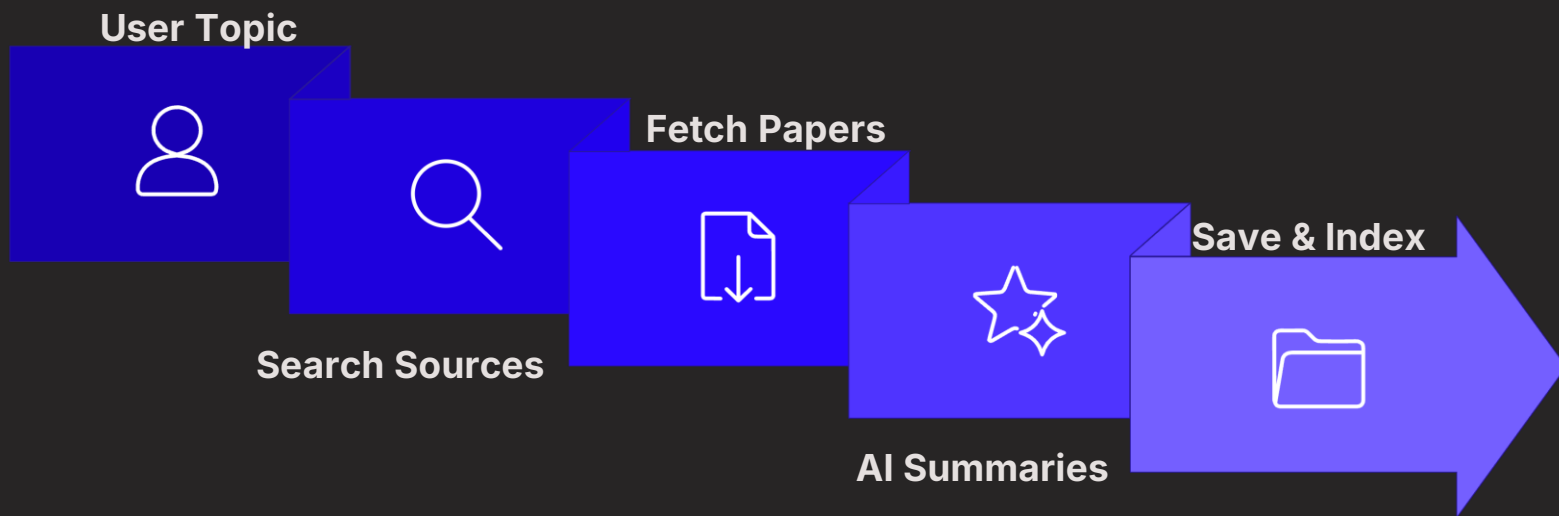


Publication notifications

Real-time alerts for new relevant papers



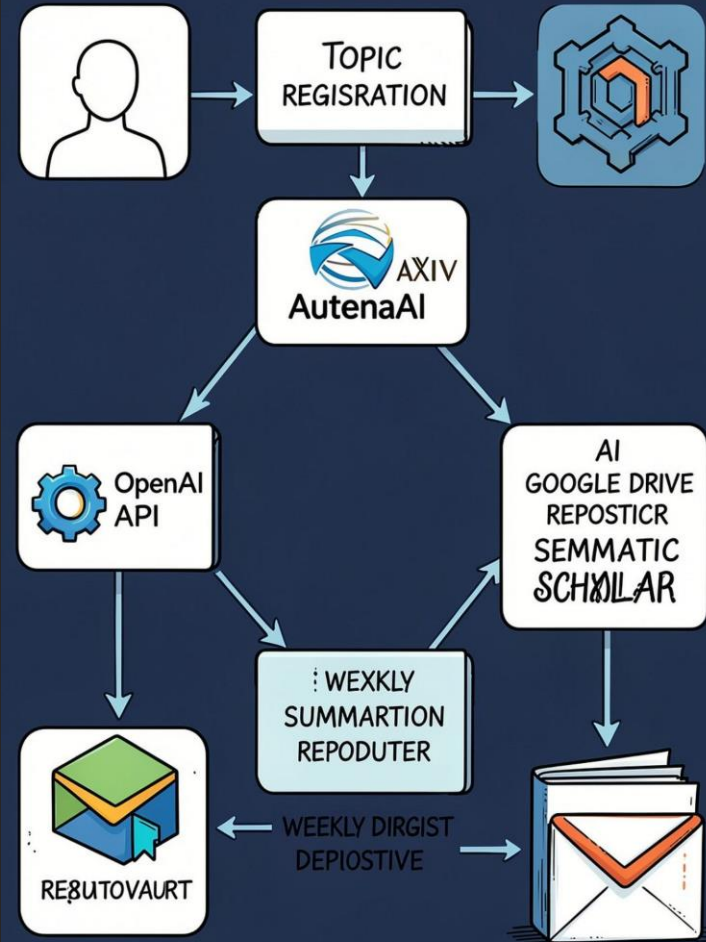
Proposed Solution — How It Works



System Architecture

Major components:

- User → Topic Registration
- n8n automation orchestrates API calls and workflows
- arXiv & Semantic Scholar for discovery
- OpenAI for summarization and keyword extraction
- Google Drive for storage; Weekly Digest generator; Gmail for notifications



Detailed Workflow



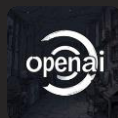
Flow highlights: automated polling, parallel API requests, lightweight metadata caching, asynchronous summarization jobs, and scheduled digest assembly.

Technologies Used



n8n

Low-code automation for orchestrating data pipelines and webhooks.



OpenAI

State-of-the-art summarization and semantic extraction models.



Google Drive

Cloud storage for PDFs, metadata, and exported collections.



Gmail

Notification channel for digests and alerts.



arXiv API

Primary source for preprints and research metadata.



Semantic Scholar API

Complementary discovery with citation graphs and influence signals.



JSON / REST APIs

Data interchange format; webhooks enable real-time updates.



Webhooks

Event-driven triggers for pipeline automation and alerting.

Key Features

AI Paper Summaries

Short, structured abstracts with key findings and methods.

Smart Keyword Extraction

Concept-level tags and ontologies for better search.

Automatic Reference Management

PDFs, bib entries, and folders synced to Google Drive.

Research Repository

Centralized, searchable storage with metadata filters.

Publication Recommendations

Personalized suggestions based on topic and citation network.

Weekly Research Digest

Curated highlights and briefs delivered via email.

Gmail Alerts

Immediate notifications for high-relevance new publications.

Searchable Database

Full-text and metadata queries with filtering and sorting.

Benefits & Applications



Saves research time

Fast triage of literature shortens discovery-to-experiment cycles.



Reduces manual effort

Automation minimizes repetitive metadata and file handling tasks.



Organizes references

Consistent structure and citations simplify writing and collaboration.



Improves productivity

Researchers focus on analysis rather than administration.



Students

Faster literature reviews and project preparation.



Faculty

Stay current with minimal overhead; support student supervision.



Researchers & Orgs

Streamline discovery, reproducibility, and knowledge sharing.



Universities

Institution-wide repository and digest capabilities.

Challenges & Future Scope

Challenges

- API rate limits and quota management across data sources
- Ensuring research data accuracy and semantic alignment
- Scalable storage, versioning, and access controls

Future Scope

**Zotero &
Mendeley
integration**

**Automated
citation &
reference
generator**

**Multi-language
summarization &
PDF annotation**

**AI chat interface
for interactive
paper Q&A**

Personalized recommendation engine

Conclusion

Project Impact

- Complete AI-powered research workflow
- Reduced manual effort & time
- Improved research productivity
- Smart paper organization
- Better decision-making with AI summaries



Future Enhancements

- Zotero & Mendeley integration
- AI Chat for research papers
- Automatic citation generator
- Multi-language summarization
- Personalized recommendation engine

Thank You - AI Research Assistant Platform | Empowering Smarter Research Through AI